

Classifier →	Logistic regression (base method)				KNN ($n_neighbors = 5$)				CatBoost (default parameters)			
Multi-Domain												
Train ↓ / Eval →	ArXiv	XSum	Writing	Review	ArXiv	XSum	Writing	Review	ArXiv	XSum	Writing	Review
ArXiv	100.00	97.34	86.82	94.05	100.00	50.25	50.00	50.50	100.00	96.61	92.39	96.65
XSum	97.80	99.94	83.02	95.24	87.13	99.65	72.37	84.70	97.52	99.93	83.21	94.42
Writing	86.74	81.82	99.98	99.40	77.89	63.40	99.85	97.97	85.53	78.78	99.99	99.35
Review	94.17	93.96	99.38	99.96	53.90	64.96	80.85	99.74	93.85	91.24	99.10	99.96

Table 10: Ablation on the downstream classifier on DetectRL Multi-Domain, per-cell AUROC (rows = training subset, columns = evaluation subset).

Steering vector construction →	Logistic regression (base method)				Mean difference				PCA			
Multi-Domain												
Train ↓ / Eval →	ArXiv		XSum		Writing	Review	ArXiv		XSum		Writing	Review
ArXiv	100.00	97.34	86.82	94.05			100.00	77.44	71.93	79.29	91.53	83.37
XSum	97.80	99.94	83.02	95.24			91.02	99.54	79.25	88.26	69.20	96.20
Writing	86.74	81.82	99.98	99.40			85.27	90.26	99.06	97.16	76.18	82.37
Review	94.17	93.96	99.38	99.96			89.63	87.84	98.14	99.40	83.72	84.66

Table 11: Ablation on the steering vector construction method on DetectRL Multi-Domain, per-cell AUROC (rows = training subset, columns = evaluation subset).

DetectRL test config	Zero-shot			RoBERTa (matched: MIRAGE polish-500)		SV-Detect variants (matched: MIRAGE)		
	Fast-DetectGPT	Binoculars	RADAR	RoBERTa-Base	RoBERTa-Large	polish-500 (default)	1-vec (union-800)	3-vec (task-aware)
<i>Multi-Domain</i>								
ArXiv	0.437	0.840	0.950	0.905	0.873	0.972	0.988	0.983
XSum	0.392	0.774	0.998	0.891	0.905	0.960	0.964	0.963
Writing	0.742	0.944	0.793	0.798	0.857	0.919	0.930	0.912
Review	0.770	0.900	0.896	0.774	0.835	0.918	0.879	0.925
Mean	0.585	0.865	0.909	0.842	0.867	0.942	0.940	0.946
<i>Multi-LLM</i>								
GPT-3.5	0.656	0.881	0.946	0.977	0.984	0.991	0.990	0.993
Claude	0.300	0.552	0.811	0.692	0.693	0.884	0.900	0.867
PaLM-2	0.660	0.933	0.951	0.786	0.813	0.902	0.918	0.874
Llama-2	0.768	0.966	0.968	0.919	0.951	0.976	0.968	0.974
Mean	0.596	0.833	0.919	0.844	0.860	0.938	0.944	0.927
<i>Multi-Attack (mean over sub-attacks)</i>								
Direct	0.796	0.949	0.922	0.907	0.912	0.973	0.973	0.970
Prompt	0.784	0.935	0.887	0.902	0.907	0.960	0.959	0.957
Paraphrase	0.701	0.883	0.943	0.818	0.833	0.939	0.944	0.940
Perturbation	0.496	0.769	0.922	0.832	0.856	0.923	0.934	0.909
Mixing	0.672	0.891	0.935	0.885	0.875	0.952	0.956	0.944
Mean	0.690	0.885	0.922	0.869	0.876	0.949	0.953	0.944
Grand mean (13 configs)	0.629	0.863	0.917	0.852	0.866	0.943	0.947	0.940

Table 12: MIRAGE → DetectRL cross-benchmark transfer, per-cell AUROC. Multi-Attack rows are unweighted means over the DetectRL sub-attack categories (Direct: 1 test set; Prompt: 3 sub-attacks; Paraphrase / Perturbation: 4 each; Mixing: 3).

MIRAGE test setting	Zero-shot			RoBERTa (matched: DetectRL joint)		SV-Detect variants (matched: DetectRL)	
	Fast-DetectGPT	Binoculars	RADAR	RoBERTa-Base	RoBERTa-Large	1-vec (default)	4-vec (task-aware)
DIG-Generate	0.777	0.890	0.865	0.928	0.964	0.975	0.952
DIG-Polish	0.572	0.624	0.657	0.844	0.897	0.859	0.831
DIG-Rewrite	0.546	0.608	0.685	0.839	0.894	0.852	0.821
SIG-Generate	0.771	0.890	0.868	0.924	0.964	0.975	0.946
SIG-Polish	0.573	0.620	0.649	0.848	0.898	0.857	0.834
SIG-Rewrite	0.548	0.604	0.681	0.836	0.890	0.846	0.816
Mean	0.631	0.706	0.734	0.870	0.918	0.894	0.867

Table 13: DetectRL → MIRAGE reverse transfer, per-cell AUROC across the six MIRAGE test settings. Bold marks the winning method per row.

Detector	RAID TPR@FPR=5%			PADBen sentence-pair AUROC per task					
	Chat	Non-Chat	Overall	PSA	GAD	AIT-L	IPD	DPA	Mean
Zero-shot baselines on the target benchmark									
LLMDet	37.4	27.6	32.5	–	–	–	–	–	–
GLTR	69.9	50.4	60.1	0.429	0.436	0.401	0.514	0.488	0.454
Fast-DetectGPT	77.5	58.4	67.9	0.638	0.787	0.504	0.478	0.369	0.555
RADAR	79.5	58.9	69.2	0.728	0.910	0.748	0.526	0.909	0.764
Binoculars (Falcon-7B)	86.7	58.8	72.8	0.399	0.457	0.505	0.498	0.457	0.463
Ours (default single-direction)									
SV-Detect (MIRAGE polish-500)	71.4	10.2	40.8	0.890	0.943	0.778	0.498	0.949	0.812
SV-Detect (DetectRL 1-vec)	65.4	26.7	46.1	0.944	0.968	0.790	0.524	0.978	0.841
Ours (multi-direction extensions)									
SV-Detect (MIRAGE 1-vec union-800)	71.7	9.4	40.5	0.886	0.942	0.753	0.494	0.939	0.803
SV-Detect (MIRAGE 3-vec task-aware)	73.9	13.2	43.6	0.931	0.952	0.762	0.503	0.969	0.823
SV-Detect (DetectRL 4-vec task-aware)	78.9	33.7	56.3	0.876	0.948	0.782	0.509	0.950	0.813

Table 14: Full zero-shot cross-benchmark results. **RAID:** TPR@FPR=5%, macro-averaged over the 8 open-source configurations, split into Chat (llama-chat, mistral-chat, mpt-chat) and Non-Chat (gpt2, mistral, mpt) LLM families. **PADBen:** sentence-pair AUROC per task (PSA = paraphrase-source attribution, GAD = general authorship detection, AIT-L = AI-text laundering, IPD = iterative paraphrase depth, DPA = deep-paraphrase attack)

G Additional comparison with RepreGuard

RepreGuard (Chen et al., 2025) is the closest representation-based AI-generated text detection method to SV-Detect.

Table 7 reports results of RepreGuard under our DetectRL evaluation protocol using the GPT-Neo-2.7B backbone. However, the default backbone of RepreGuard is LLaMA-3.1-8B. In this section, we provide an additional comparison of SV-Detect to RepreGuard under the matched backbones using the default setup of Chen et al. (2025): 512-pair train/test cells with per-LLM stratification, and macro-averaging across a small number of bootstrap seeds. We use RAID (Dugan et al., 2024) for comparison, which is the default benchmark used in Chen et al. (2025).

Table 15 reports RAID performance for RepreGuard and SV-Detect trained on RAID official training split, under RepreGuard’s training protocol, at both GPT-Neo-2.7B and LLaMA-3.1-8B-Instruct. We follow the aggregation of Dugan et al. (2024) and macro-average each row across the eight open-source RAID configurations (Chat vs. Non-Chat $\times$ greedy vs. sampling $\times$ repetition-penalty on/off).

Results suggest that RepreGuard is highly backbone-sensitive: at GPT-Neo-2.7B its RAID score collapses to near-chance (Chat 4.5% TPR@5%, Non-Chat 5.3%; AUROC $\approx$ 0.5), and only moving to LLaMA-3.1-8B-Instruct recovers Chat performance (92.9% TPR@5%). Non-Chat generators remain hard for RepreGuard even at LLaMA-8B (41.7%). The same backbone-sensitivity is visible on DetectRL Multi-LLM: our reproduction of RepreGuard under its own protocol yields 2.5% TPR@5% (AUROC 0.366) at GPT-Neo-2.7B and 30.3% TPR@5% (AUROC 0.812) at LLaMA-3.1-8B-Instruct, macro-averaged across the four Multi-LLM generators. SV-Detect does not exhibit this dependence: at both GPT-Neo-2.7B and LLaMA-3.1-8B it reaches $\approx$ 99.7% TPR@5% / $\approx$ 1.0 AUROC on both Chat and Non-Chat RAID cells. In other words, SV-Detect matches or beats RepreGuard at its *best* backbone while remaining strong at the small GPT-Neo backbone where RepreGuard collapses. This supports our claim that the LogReg-based steering construction produces a discriminative direction that is robust to backbone choice, whereas RepreGuard’s cluster-separation heuristic degrades sharply when

Method	Backbone	TPR@5%	AUROC
<i>Zero-shot / score-based (RAID Table 5)</i>			
LLMDet	–	32.5	–
GLTR	–	60.1	–
Fast-DetectGPT	–	67.9	–
RADAR	–	69.2	–
Binoculars	Falcon-7B	72.8	–
<i>Supervised, applied zero-shot to RAID (RAID Table 5)</i>			
RoBERTa-Base (GPT-2)	–	59.9	–
<i>Trained on RAID</i>			
RepreGuard	GPT-Neo-2.7B	4.9	0.5500
RepreGuard	LLaMA-3.1-8B	93.2	0.9634
SV-Detect (ours)	GPT-Neo-2.7B	99.7	0.9993
SV-Detect (ours)	LLaMA-3.1-8B	99.7	0.9993

Table 15: RAID performance on the eight open-source configurations (Chat / Non-Chat $\times$ decoding $\times$ repetition penalty).

the observer LLM’s representations are less well-separated.